CSE331: Microprocessor Interfacing and Embedded Systems

Lecture#11: Interrupt Part#1 (Chapter#11)

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Interrupts

- Motivations
  - Inform a program of some external events timely
    - Polling *vs* Interrupt
  - Implement multi-tasking with priority support

Merriam-Webster:

"to break the uniformity or continuity of"

Polling *vs* Interrupt

Polling:
You pick up the phone every three seconds to check whether you are getting a call.

Interrupt:
Do whatever you should do and pick up the phone when it rings.

Image from http://vecto.rs

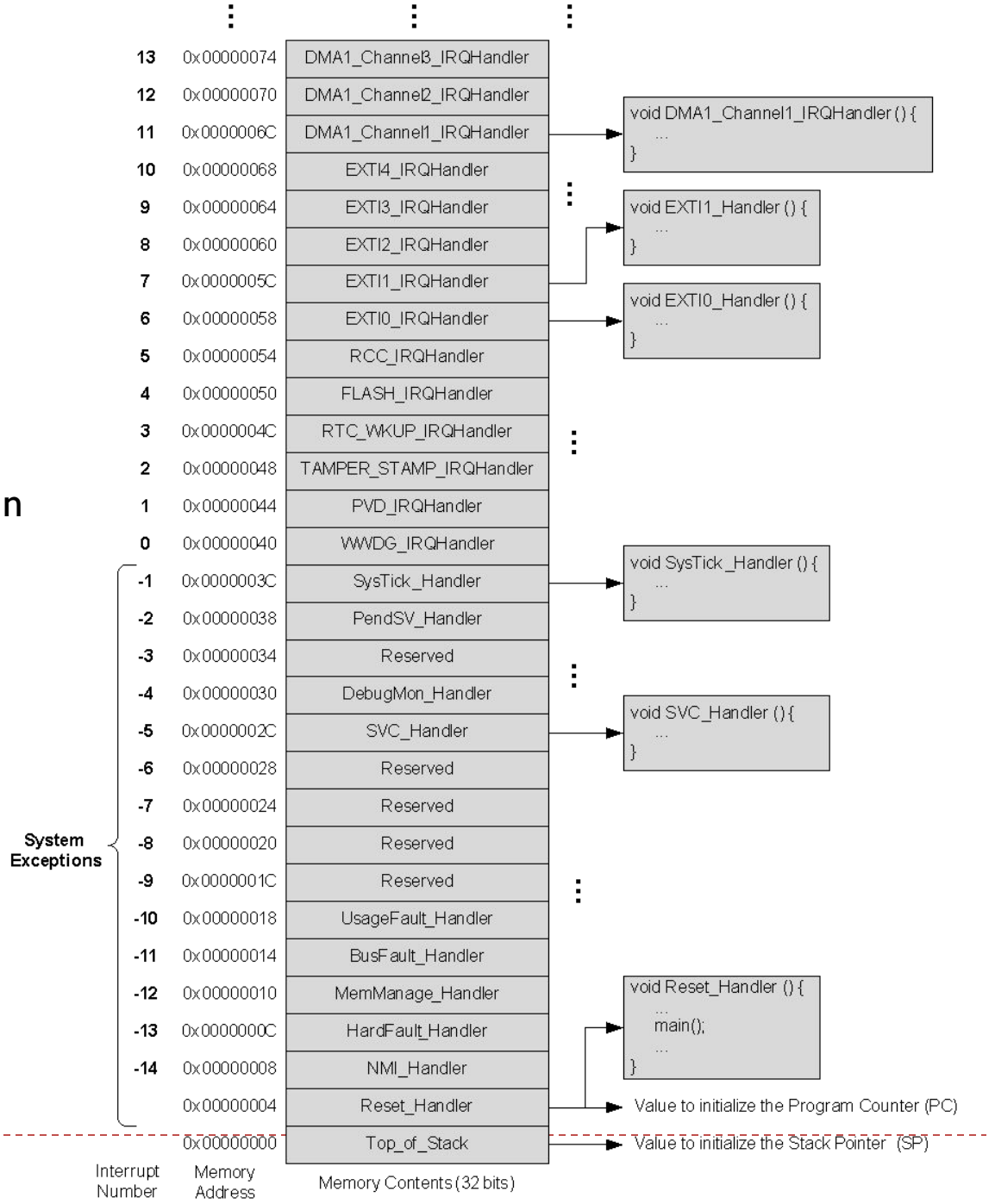
Interrupt Service Routine Vector Table

- Start address for the exception hander for each exception type is fixed and pre-defined
- Processor loads PC with this fixed, pre-defined address
- Exception Vector Table starts at memory address 0
- Program Counter pc = 0x00000004 initially

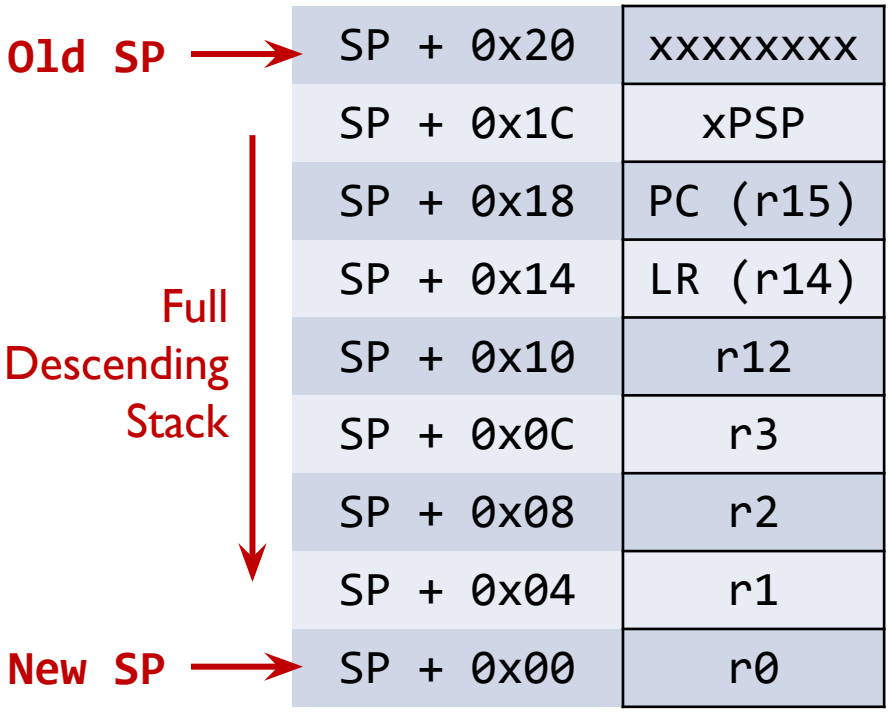
Address	Priority	Type of priority	Acronym	Description
0x0000_0000	-	-	-	Stack Pointer
0x0000_0004	-3	fixed	Reset	Reset Vector
0x0000_0008	-2	fixed	NMI_Handler	Non maskable interrupt. The RCC Clock Security System (CSS) is linked to the NMI vector.
0x0000_000C	-1	fixed	HardFault_Handler	All class of fault
0x0000_0010	0	settable	MemManage_Handler	Memory management
0x0000_0014	1	settable	BusFault_Handler	Pre-fetch fault, memory access fault
0x0000_0018	2	settable	UsageFault_Handler	Undefined instruction or illegal state
0x0000_001C-0x0000_002B	-	-	-	Reserved
0x0000_002C	3	settable	SVC_Handler	System service call via SWI instruction
0x0000_0030	4	settable	DebugMon_Handler	Debug Monitor
0x0000_0034	-	-	-	Reserved
0x0000_0038	5	settable	PendSV_Handler	Pendable request for system service
0x0000_003C	6	settable	SysTick_Handler	System tick timer
...				

ISR Vector Table

For interrupt number n:
CMSIS Interrupt Number = 16 + n



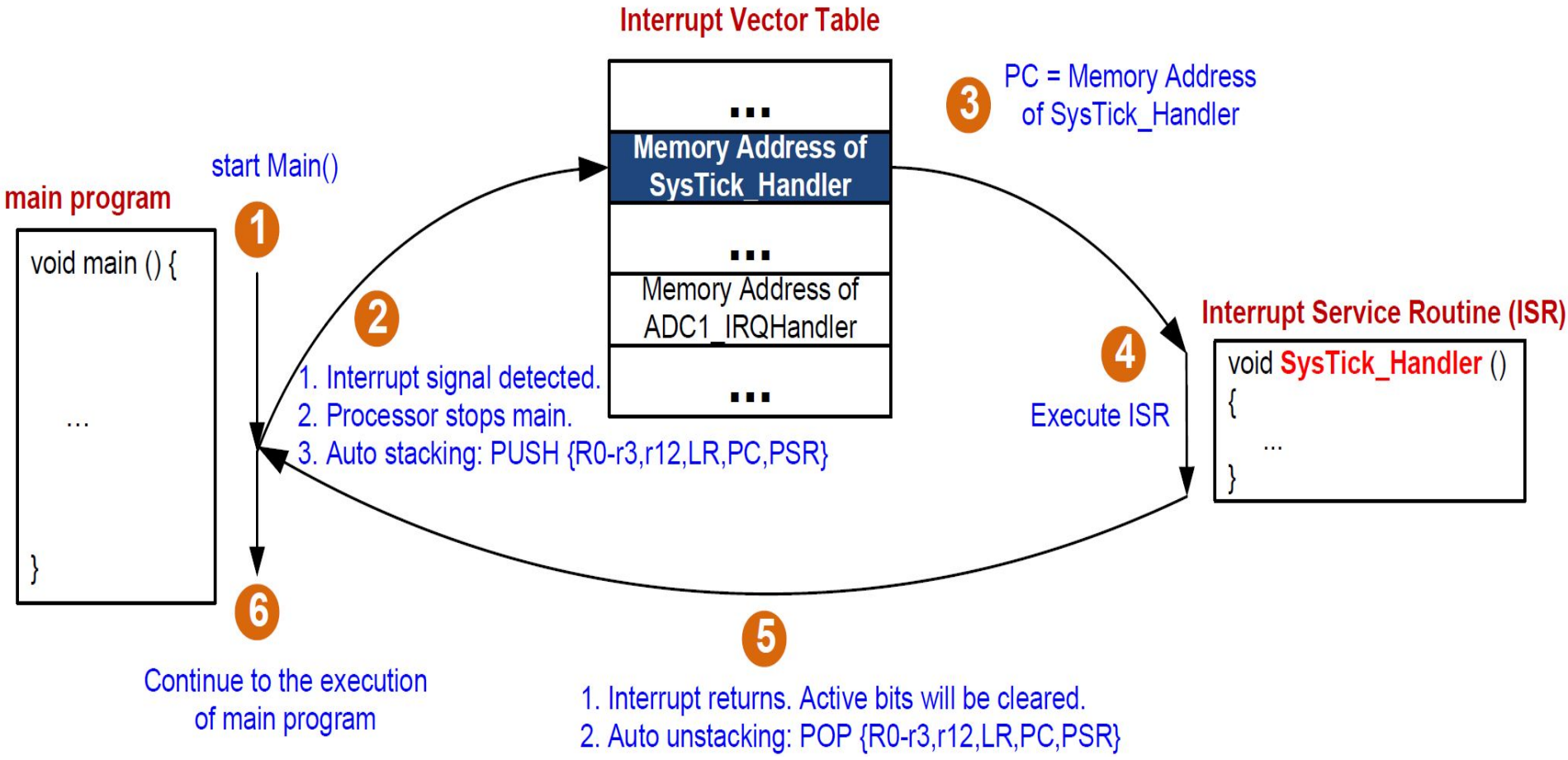
Stacking & Unstacking



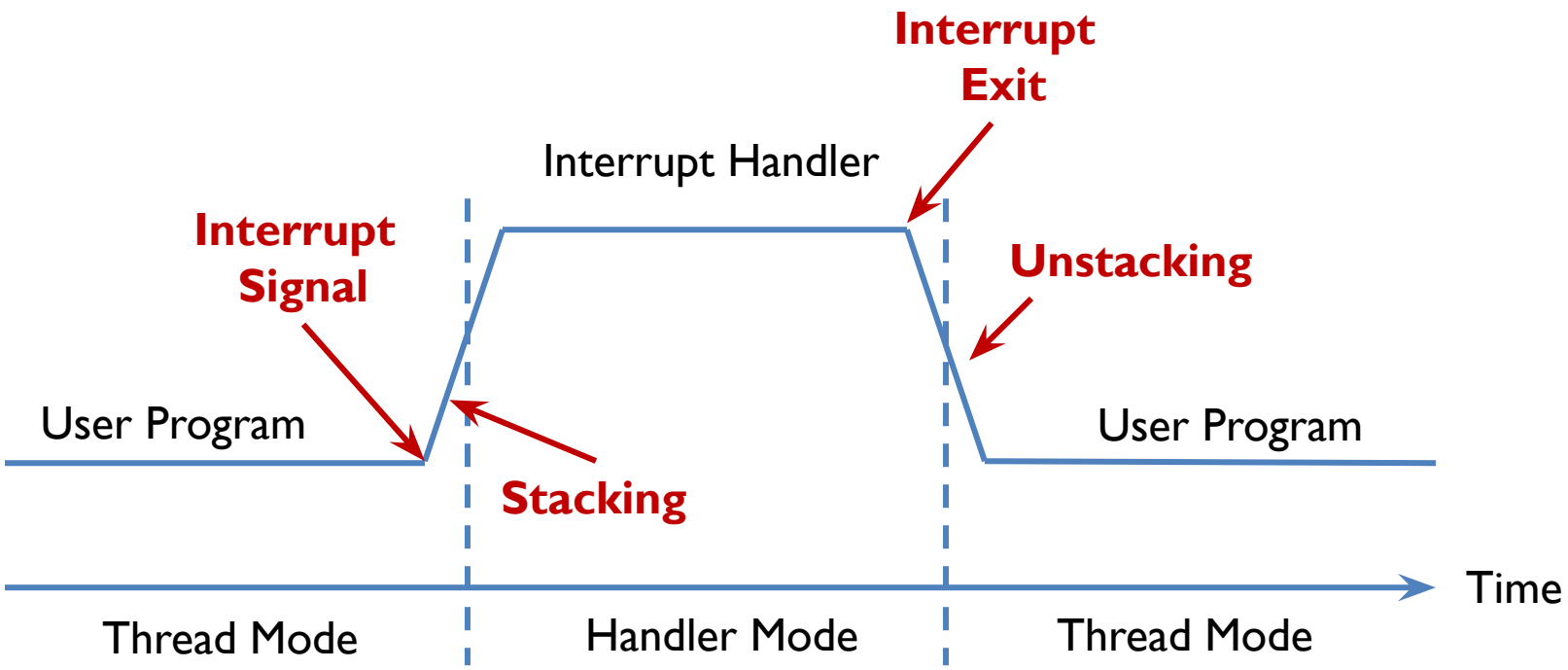
- **Stacking:** The processor automatically pushes these eight registers into the main stack before an interrupt handler starts
- **Unstacking:** The processor automatically pops these eight register out of the main stack when an interrupt hander exits.

- Two stack pointers: Main SP (MSP) and Process SP (PSP)
- Determined by operating mode, and bit 0 of the CONTROL register
  - Handler mode ⟶ SP = MSP
  - Thread mode ⟶ SP = MSP, if CONTROL[0] = 0 (i.e., privileged thread mode);
    SP = PSP, if CONTROL[0] = 1 (i.e., unprivileged thread mode)

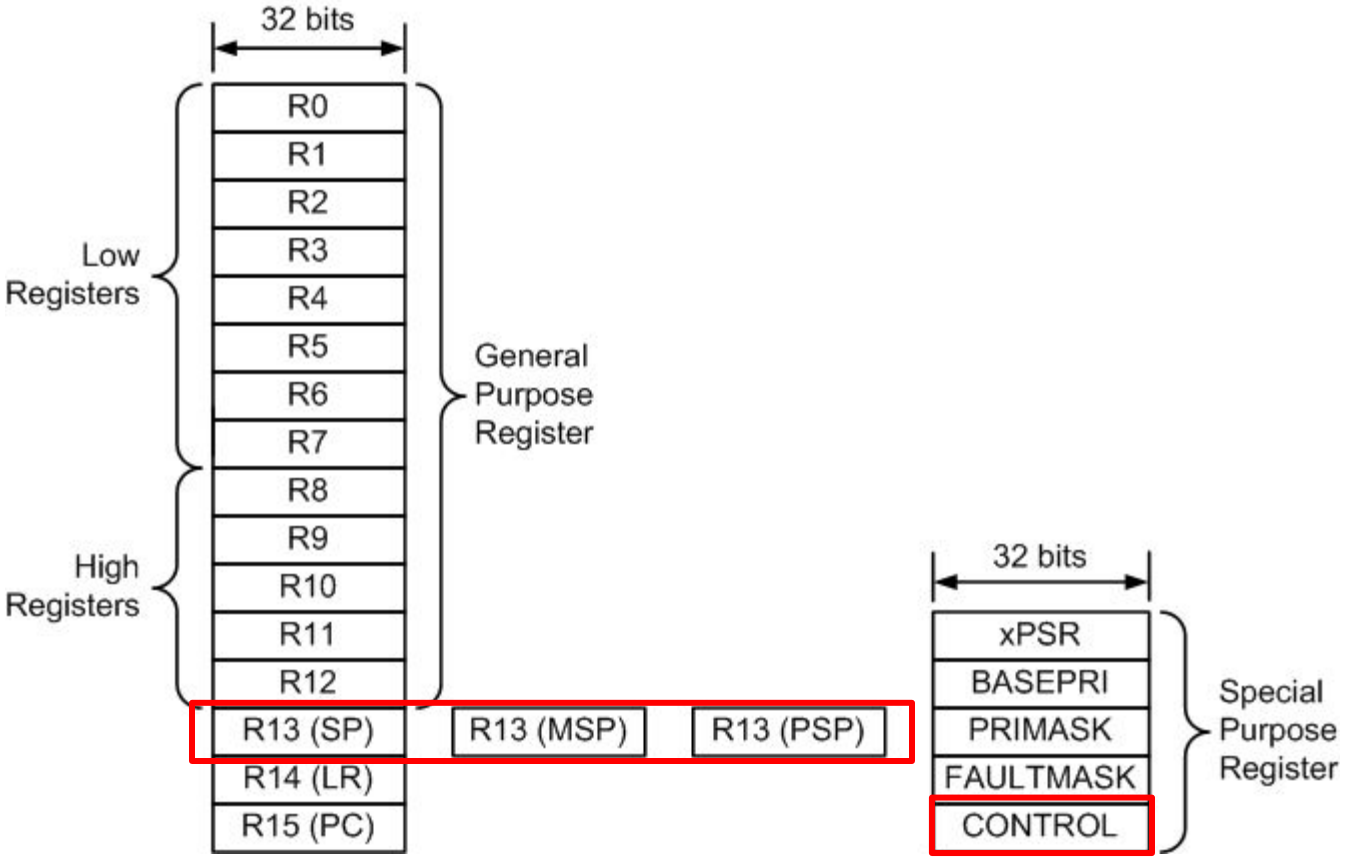
Interrupt


Interrupt Vector Table
...
Memory Address of SysTick_Handler
...
Memory Address of ADC1_IRQHandler
...
3
PC = Memory Address of SysTick_Handler
start Main()
main program
1
void main () {
...
}
2
1. Interrupt signal detected.
2. Processor stops main.
3. Auto stacking: PUSH {R0-r3,r12,LR,PC,PSR}
4
Execute ISR
Interrupt Service Routine (ISR)
void SysTick_Handler ()
{
...
}
5
1. Interrupt returns. Active bits will be cleared.
2. Auto unstacking: POP {R0-r3,r12,LR,PC,PSR}
6
Continue to the execution of main program

Stacking & Unstacking



Registers



MSP: Main Stack Pointer

PSP: Process Stack Pointer

Processor Mode: Handler Mode *vs* Thread Mode

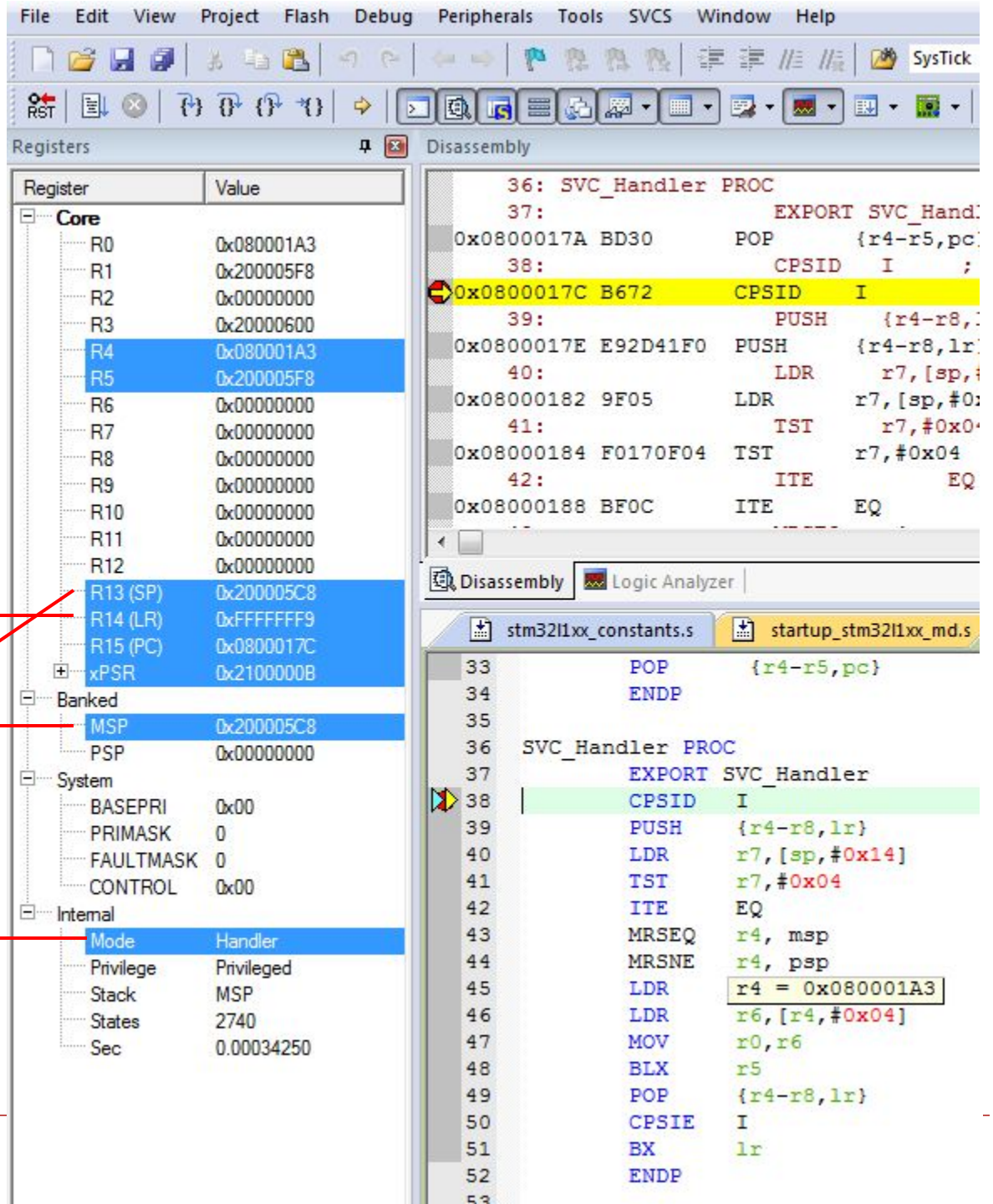
- Handler mode and Thread mode
  - Handler mode always use MSP (Main Stack Pointer)
  - Thread Mode uses either PSP (Process Stack Pointer) or MSP
    - Control[1] = 0, SP = MSP (default)
    - Control[1] = 1, SP = PSP
- When the processor is reset, the default is the thread mode.
- The processor enters the handler mode when an exception occurs.

Register values in a interrupt service routine

LR = 0xFFFFFFF9

SP = MSP

ISR always in hander mode.



Which stack to use when exiting an interrupt?

Link Register (LR) now has two usages:

- LR = address of the instruction immediately after BL
- LR indicates whether MSP or PSP is used to restore register from when exiting an interrupt
  - LR value generated by processor
  - The processor set LR to **0xFFFFFFFF** on reset.
  - If return to handler mode, the MSP stack is always used

Which stack to use when exiting an interrupt?

Link Register (LR) now has two usages:

- LR = address of the instruction immediately after BL
- LR indicates whether MSP or PSP is used to restore register from when exiting an interrupt

No FP extension:

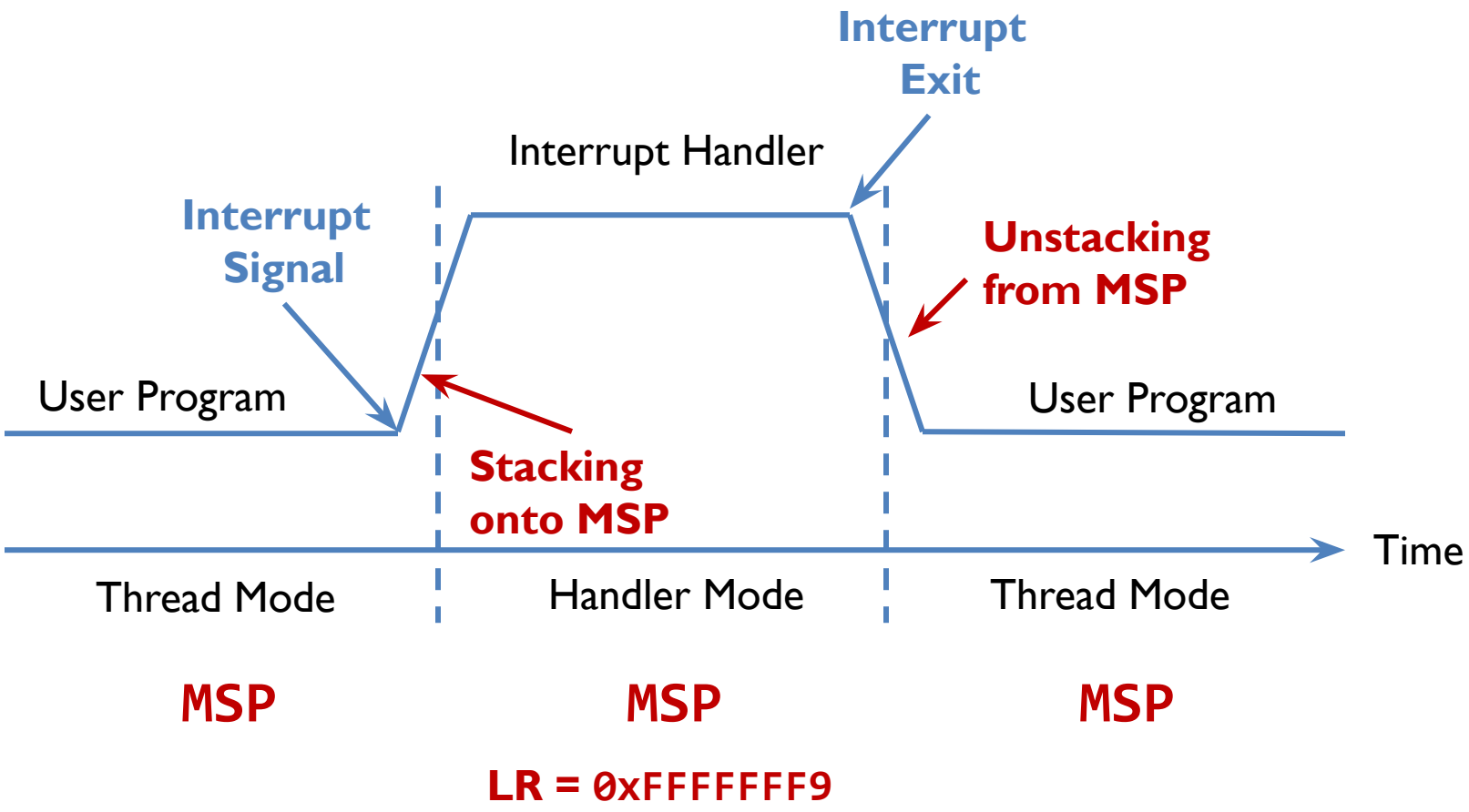
Link Register (LR)	Return Mode	Return Stack
0xFFFFFFF1	Handler	SP = MSP
0xFFFFFFF9	Thread	SP = MSP
0xFFFFFFFD	Thread	SP = PSP

With FP extension:

Link Register (LR)	Return Mode	Return Stack
0xFFFFFFE1	Handler	SP = MSP
0xFFFFFFE9	Thread	SP = MSP
0xFFFFFFED	Thread	SP = PSP

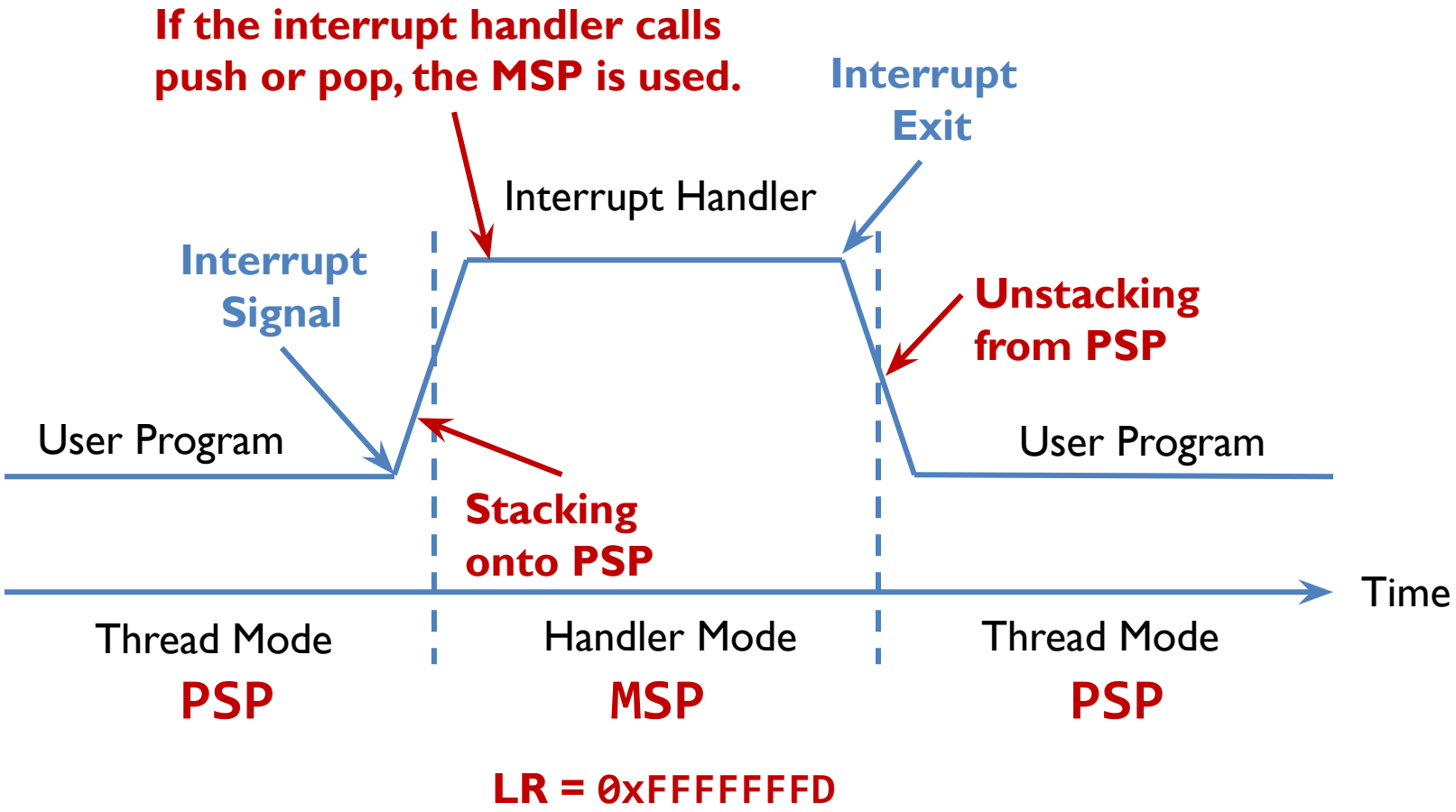
Stacking & Unstacking

Control[1] = 0 ⟹ User program uses MSP.



Stacking & Unstacking

Control[1] = 1 ⟹ User program uses PSP.



LR = 0xFFFFFFFD

Which is being used?

When exiting an interrupt:

- If LR = 0xFFFFFFF9, then SP = MSP
- If LR = 0xFFFFFFFD, then SP = PSP

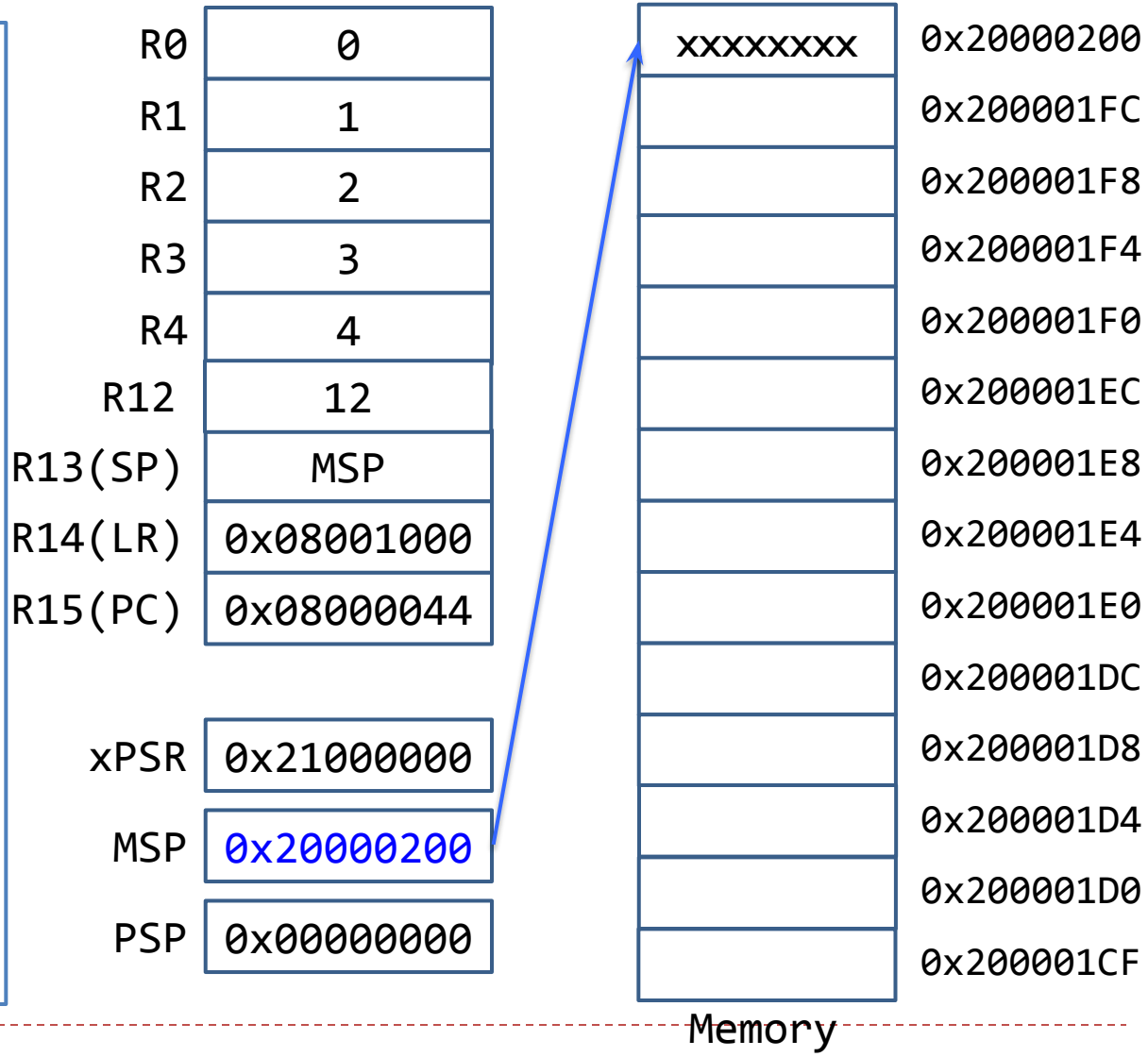
9 = 1001

D = 1101

```
TST   r7, #0x04
MRSEQ r4, msp
MRSNE r4, psp
STMFD r4!, {r7-r9}  ; Push onto a Full Descending Stack
```

Interrupt: Stacking & Unstacking

```
__main PROC
                addr = 0x08000044
   …
   MOV r3,#0
   …
   ENDP
                addr = 0x0800001C
SysTick_Handler PROC
   EXPORT SysTick_Handler
   ADD r3, #1
   ADD r4, #1
   BX  lr
   ENDP
```



Interrupt: S

Suppose SysTick interrupt occurs when PC = 0x08000044

```
__main PROC
   …          addr = 0x08000044
   MOV r3,#0
   …
   ENDP
              addr = 0x0800001C
SysTick_Handler PROC
   EXPORT SysTick_Handler
   ADD r3, #1
   ADD r4, #1
   BX  lr
   ENDP
```

R0	0
R1	1
R2	2
R3	3
R4	4
R12	12
R13(SP)	MSP
R14(LR)	0x08001000
R15(PC)	0x08000044
xPSR	0x21000000
MSP	0x20000200
PSP	0x00000000

xxxxxxxx	0x20000200
	0x200001FC
	0x200001F8
	0x200001F4
	0x200001F0
	0x200001EC
	0x200001E8
	0x200001E4
	0x200001E0
	0x200001DC
	0x200001D8
	0x200001D4
	0x200001D0
	0x200001CF

Memory

Interrupt: Stacking & Uns

STACKING

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```

R0	0
R1	1
R2	2
R3	3
R4	4
R12	12
R13(SP)	MSP
R14(LR)	0xFFFFFFF9
R15(PC)	0x0800001C

xPSR	0x21000000
MSP	0x200001E0
PSP	0x00000000

	xxxxxxxx	0x20000200
xPSR	0x21000000	0x200001FC
PC	0x08000044	0x200001F8
LR	0x08001000	0x200001F4
R12	12	0x200001F0
R3	3	0x200001EC
R2	2	0x200001E8
R1	1	0x200001E4
R0	0	0x200001E0
		0x200001DC
		0x200001D8
		0x200001D4
		0x200001D0
		0x200001CF

Memory

Interrupt: Stacking & Uns

STACKING

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__main PROC
   …
   MOV r3,#0
   …
   ENDP

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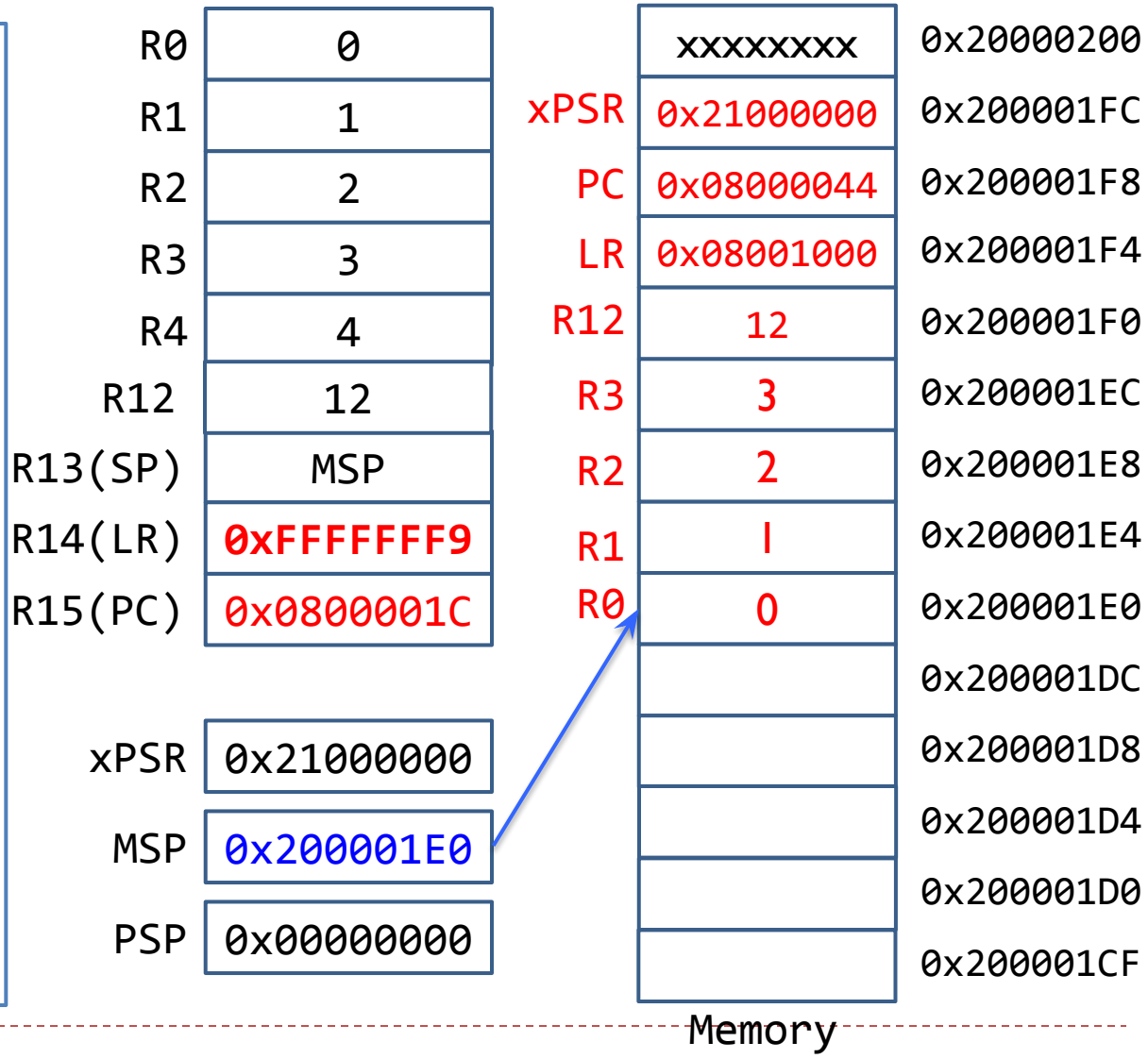
```

addr = 0x08000044

addr = 0x0800001C

LR = 0xFFFFFFF9 to indicate MSP is used.



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```

R0	0
R1	1
R2	2
R3	**4**
R4	4
R12	12
R13(SP)	MSP
R14(LR)	0xFFFFFFF9
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xPSR	0x21000000
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LR	0x08001000	0x200001F4
R12	12	0x200001F0
R3	3	0x200001EC
R2	2	0x200001E8
R1	1	0x200001E4
R0	0	0x200001E0
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Memory

Interrupt: Stacking & Unstacking

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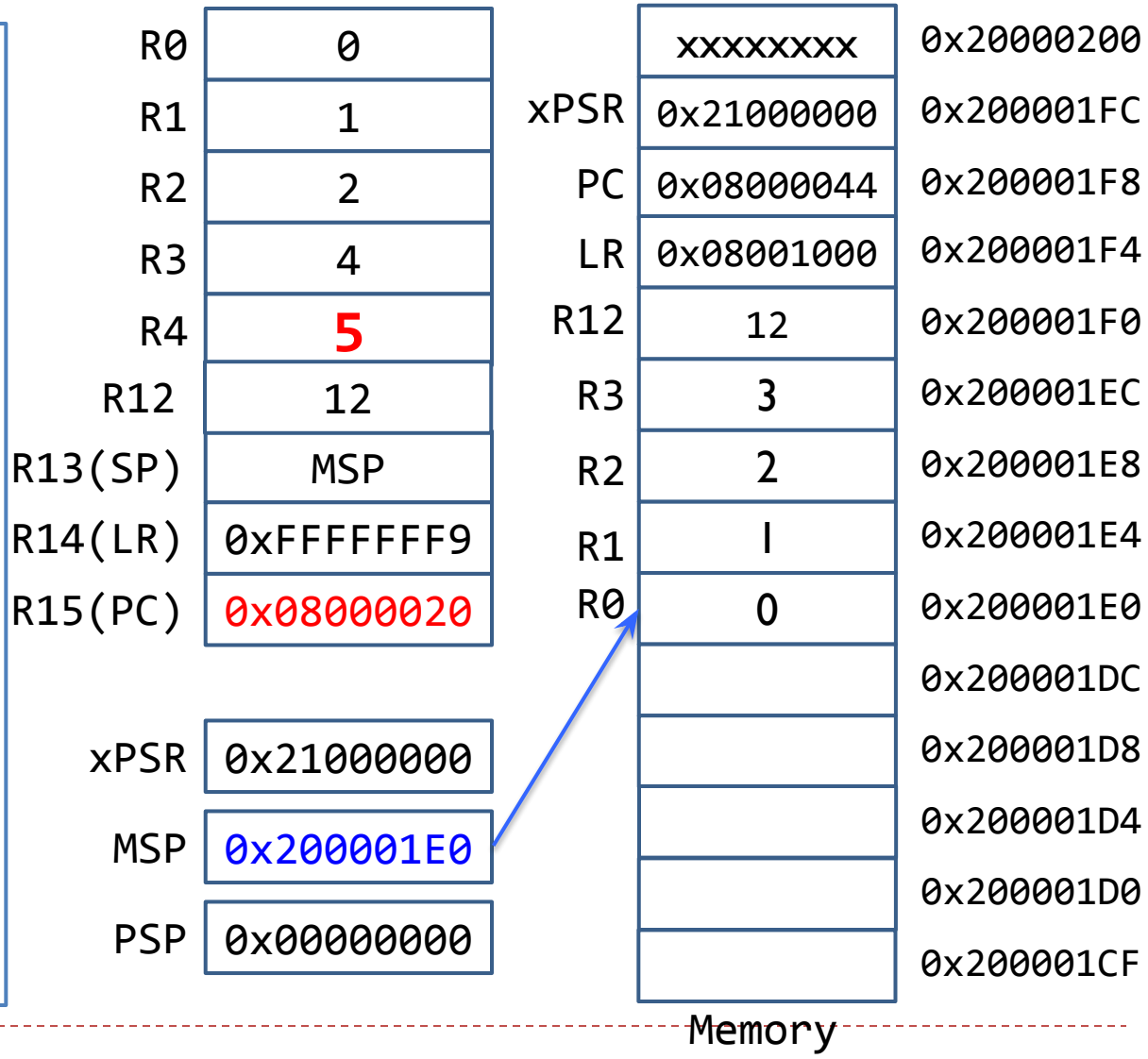

Interrupt: Stacking & Unstacking

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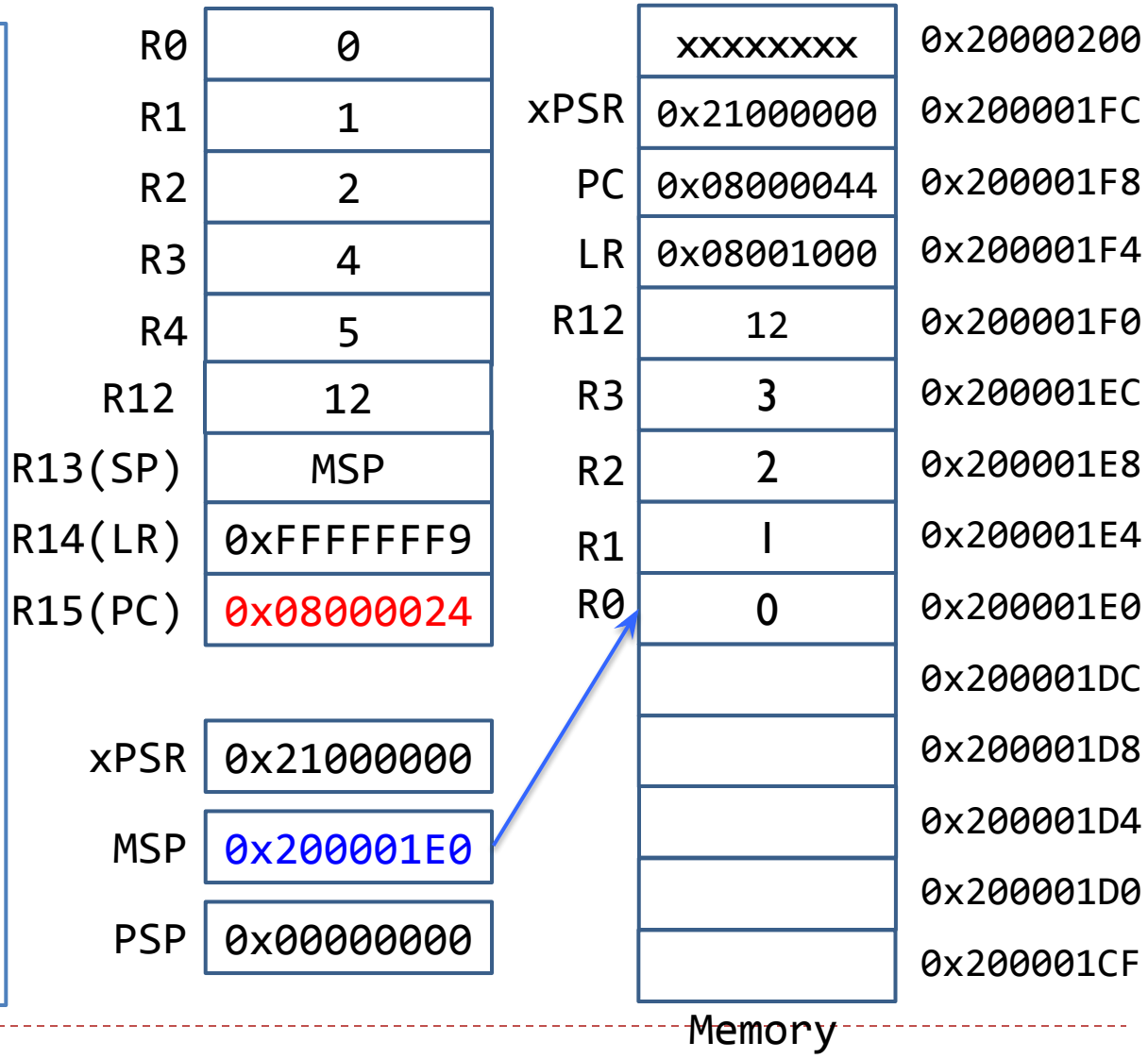

Interrupt: Stacking & Uns UNSTACKING

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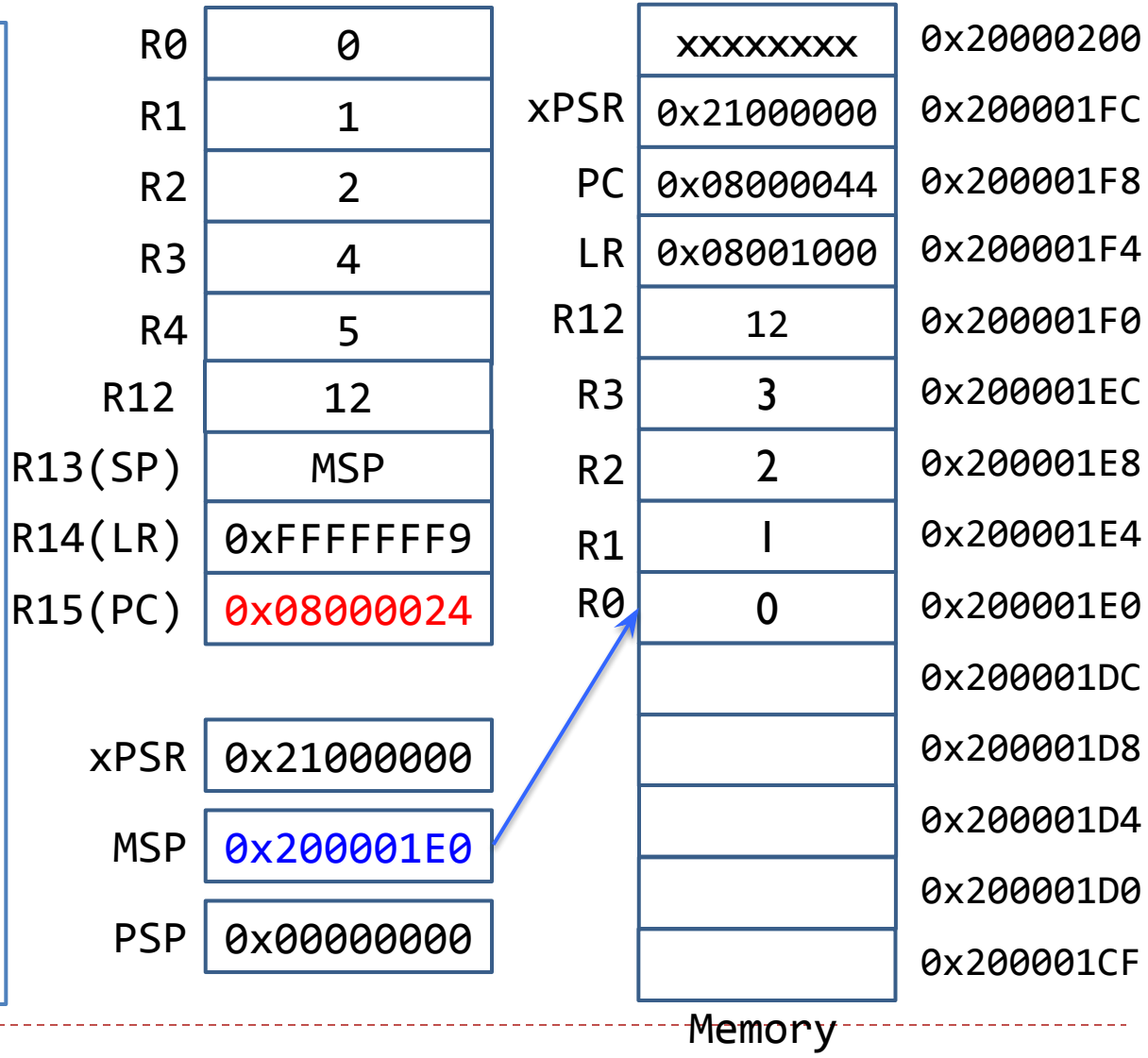

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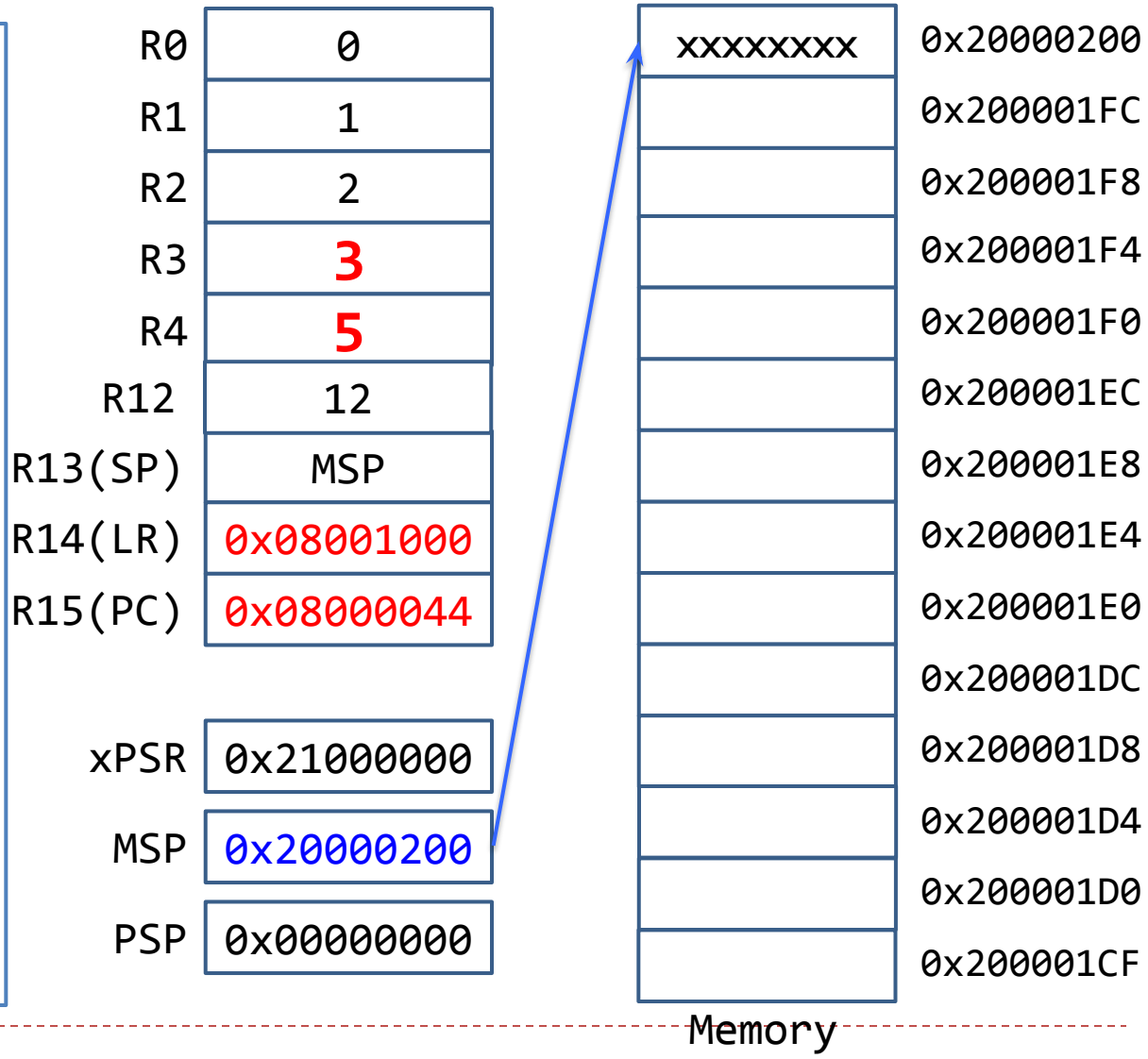
   ENDP
```

Note the new value of R3 is lost!!!



Interrupt: Stacking & Unstacking

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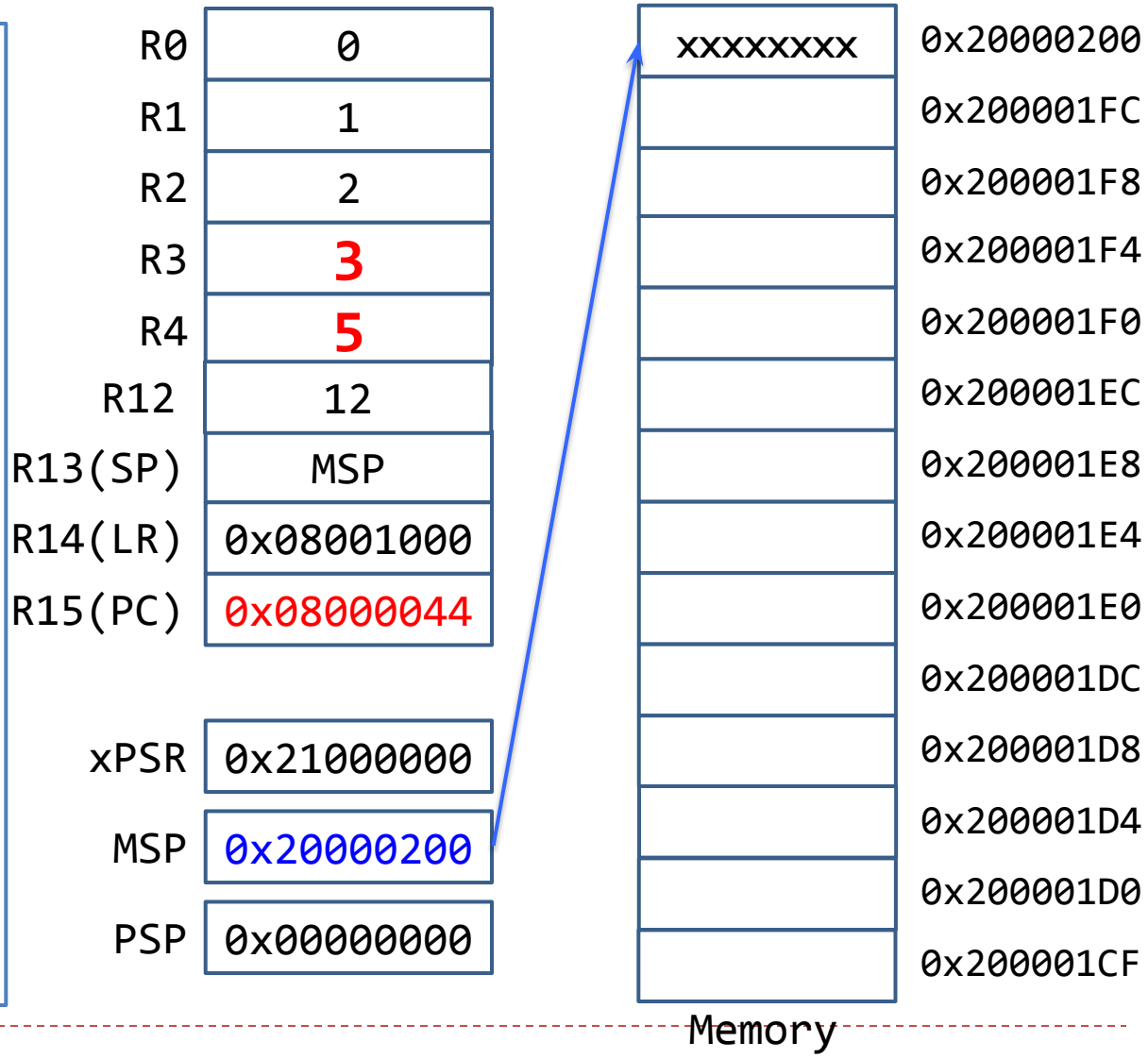
```

The Main program resumes!!!



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